

April 24, 2023

AGWET User's Manual

Version v6.00

Document Revision:

April 24, 2023

For use beginning with
TopAGNPS v6.00

Table of Contents:

Table of Contents:	3
AGWET User's Manual:	5
Introduction:	5
Input files:	6
The AGWET Control file:	6
AGWET.CSV or AGWET.XML	6
AGWET Control File Keywords:	7
Required spatial layer input data files:	11
AnnAGNPS_Cell_IDs.asc:	11
AnnAGNPS_Reach_IDs.asc:	11
BOUND.ASC:	11
FLOVEC.ASC:	11
RELIEF.ASC:	11
TSLOPE.ASC:	12
UPAREA.ASC:	12
Optional input data files:	12
AnnAGNPS_Cell_Data_Section.csv:	12
AgFlow_LS_Factor.asc:	12
User-specified filename (ASCII grid or comma-delimited):	12
PEG_Summary.csv:	12
AnnAGNPS_Ephemeral_Gully_Data_Section.csv:	13
AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgBuf.csv:	13
Output files:	13
AgWet.log:	13
AgWet_AnnAGNPS_Cell_Summary.csv:	13
AgWet_AnnAGNPS_Reach_Summary.csv:	14
AnnAGNPS_Riparian_Buffer_Data_Section_AgWet.csv:	14
AgWet_AnnAGNPS_Wetland_Barrier_Lines.asc:	14
AgWet_AnnAGNPS_Wetland_Barrier_Points.asc:	14
AgWet_AnnAGNPS_Wetland_Barrier_Points.csv:	14
AgWet_AnnAGNPS_Wetland_Buffer_IDs.asc:	14
AgWet_AnnAGNPS_Wetland_Buffer_Veg.asc:	14
AgWet_AnnAGNPS_Wetland_Buffer_Zone.asc:	15
AnnAGNPS_Wetland_Data_Section.csv:	15
AgWet_AnnAGNPS_Wetland_Edge.asc:	15
AgWet_AnnAGNPS_Wetland_IDs.asc:	15
AgWet_Buffer_Zone_Cell_Edge.asc:	15
AgWet_Buffer_Zone_Edge_Rasters.asc:	15
AgWet_Buffer_Zone_Reach_Edge.asc:	16
AgWet_Calculated_Erosion_Index_Values.asc:	16
AgWet_Cell-Reach_Buffer_Flow_Path_Summary.asc:	16
AgWet_Cell-Reach_Contributions_to_Buffers_by_Cell.asc:	16
AgWet_Cell-Reach_Contributions_to_Wetlands_by_Cell.asc:	16

AgWet_Cell-Reach_Contributions_to_Wetlands_by_Wetland.asc:	16
AgWet_Reach_Located_Buffers.asc:	16
AgWet_Relief_Before_Adding_Wetland_Barriers.asc:	17
AgWet_Relief_Wetland_Extent.asc:	17
AgWet_Relief_With_Wetland_Barriers.asc:	17
AgWet_Wetland_Barrier_Common_Points.csv:	17
AgWet_Wetland_Barrier_Generated_Points.asc:	17
AgWet_Wetland_Barrier_Generated_Points.csv:	17
AgWet_Wetland_Barrier_Points_Area_Ratio_Criteria_Met.csv:	17
AgWet_Wetland_Barrier_Points_Not_Used_For_AnnAGNPS.csv:	18
AgWet_Wetness_Index_Values.asc:	18
AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgWet.csv:	19
AgWet_AnnAGNPS_Reach_Summary_Revised.csv:	19
AnnAGNPS_Reach_Data_Section_Revised.csv:	19
AnnAGNPS_Cell_Data_Section_Revised.csv:	19
AnnAGNPS_Reach_IDs_Revised.asc:	19
Manual Wetland Identification:	19
Wetland Barrier lines:	20
Barrier Line File Keywords:	20
Wetland Barrier points:	21
Barrier Points File Keywords:	21
Automatic Wetland Identification:	23
AGWET Control File Keywords:	23
"Filename"	24
"Breach_Barrier"	24
"Barrier_Height_Option"	24
"Barrier_Height"	25
"Barrier_Height_Increment"	25
"Barrier_Height_Max"	25
"Wetland_ID_Option"	25
"Reach_Enhancement_Option"	26
"Erosion_Index_Option"	26
"Erosion_Index_Threshold"	27
"DA_Threshold"	27
"WI_Threshold"	27
"Min_Wetland_Ratio"	27
"Max_Wetland_Ratio"	27

AGWET User's Manual:

Introduction:

The identification and characterization of wetlands within a watershed system for use in computer simulations can be simplified or automated using advanced wetland technology. The program “AGWET” (**AG**NPS **WET**land feature) has been developed as a module for use within the suite of modules called TopAGNPS to locate and parameterize wetlands anywhere within a watershed system based on topographic and user-defined criteria to simplify the creation of wetland input parameters for use within AnnAGNPS. The goal of this program is to produce a wetland input parameter file for use within AnnAGNPS that contains information about the wetland, such as the location of wetlands, wetland area and depth of the wetland, within stream networks for evaluation of wetland impacts on runoff, sediment, nutrient and pesticide loads.

Most of the input files for AGWET, as described below, can be generated by executing the following modules: DEDNM, RASPRO, RASFOR, and AgFlow. These TopAGNPS modules produce spatial files used by AGWET, such as the spatial output file called “AgFlow_LS_Factor.asc” that contains a RUSLE length-slope value for every raster within the watershed. AGWET can be executed independently from the other TOPAGNPS modules using the command line switch ‘/7’, as described in the TOPAGNPS User Manual.

AGWET provides two modes of operation: (i) an automated GIS Wetland Identification Feature (WIF) to generate wetland locations based on user-defined criteria and (ii) user-defined (manual) identification of potential sites for constructed wetland implementation. In the user-defined mode of operation, the physical and structural characteristics of user-identified sites are calculated and recorded into a database. In the automated mode of operation, topographic analyses are performed to define locations compatible with wetland implementation. Results from the automated mode of operation can then be used as input in the user-defined mode for subsequent integration with AnnAGNPS.

When the user-defined wetland location option is selected there are two options to indicate where wetlands are to be located. The first option allows users to define an input file of x and y points and barrier heights for each wetland that falls on a reach in the watershed. The second option allows users to define a spatial input file that describes wetland barrier lines with unique integer numerical identifiers that falls on a reach.

Another feature of AGWET is the mechanism to add buffers around the wetlands produced by AGWET. The user can choose to add buffers of a specified width through the use of keywords as described below. Then the trapping efficiency of those buffers can be calculated and reported.

AGWET produces several output files that are helpful in describing the characteristics of each wetland as described below. The main output file produced by AGWET for use as the “Wetland Data” section within AnnAGNPS is “AnnAGNPS_Wetland_Data_Section.csv”. The spatial-layer “AgWet_AnnAGNPS_Wetland_IDs.asc” file indicates the location and extent of each wetland produced in the “Wetland Data” section.

Input files:

There are seven required and two optional input files, described below, necessary for the successful execution of AGWET. Of the seven required input files, the first is a control file and the remaining six are spatial layer files. The two optional input files are output from the PEG module which AGWET uses to identify PEG points that fall within wetland areas. Also, if executing TopAGNPS on a UNIX/Linux system (case sensitive), the following list is in the expected case for each file.

The AGWET Control file:

AGWET.CSV or AGWET.XML

The required control file allows the user to specify a number of parameters that affect the processing of spatial data and the determination of wetland locations. If both "AGWET.CSV" and "AGWET.XML" are found, the "AGWET.CSV" control file will be processed. A message will be produced in the log file and AGWET execution terminated if the control file is not present.

There are two possible formats that may be used; csv-formatted or xml-formatted.

"AGWET.CSV" is a csv-formatted control file. The first record in the control file is the header record containing at least one of several possible keywords that may be used. The second record is the data record containing the values for the keywords that are specified in the header record. Only two records are read, header and data; therefore, any other information contained within the control file is ignored. The control file must be named "AGWET.CSV".

The following is an Excel example of the csv-formatted control file showing a few of the keywords:

FILENAME	BARRIER_HEIGHT_OPTION	BARRIER_HEIGHT	BARRIER_HEIGHT_INCREMENT	BARRIER_HEIGHT_MAX	WETLAND_ID_OPTION	EROSION_INDEX_OPTION	EROSION_INDEX_THRESHOLD	ASC_PATH	CSV_PATH
	1	1	0.1	1	1	0	2.5	./Output_Files/ASC	./Output_Files/CSV

"AGWET.XML" is xml-formatted control file. If the xml-format is used, it must adhere strictly to the following –only one tag per line, with no spaces. The xml-formatted control file must be named "AGWET.XML". The following is an example of the xml-formatted control file showing all available keywords.

```
<TOPAGNPS>
<KEYWORD ID="ASC_PATH">./Output_Files/ASC</KEYWORD>
<KEYWORD ID="BARRIER_HEIGHT_OPTION">1</KEYWORD>
<KEYWORD ID="BARRIER_HEIGHT">1.</KEYWORD>
<KEYWORD ID="BARRIER_HEIGHT_INCREMENT">0.1</KEYWORD>
<KEYWORD ID="BARRIER_HEIGHT_MAX">1</KEYWORD>
<KEYWORD ID="BREACH_BARRIER">1</KEYWORD>
<KEYWORD ID="BUFFER_WIDTH">3</KEYWORD>
<KEYWORD ID="BUFFER_EXTENT_OPTION">1</KEYWORD>
<KEYWORD ID="BUFFER_IDS_FILENAME"></KEYWORD>
```

```

<KEYWORD ID="BUFFER_VEG_FILENAME"></KEYWORD>
<KEYWORD ID="BUFFER_ZONE_FILENAME"></KEYWORD>
<KEYWORD ID="CSV_PATH">./Output_Files/CSV</KEYWORD>
<KEYWORD ID="DA_THRESHOLD">100.</KEYWORD>
<KEYWORD ID="EROSION_INDEX_OPTION">0</KEYWORD>
<KEYWORD ID="EROSION_INDEX_THRESHOLD">2.5</KEYWORD>
<KEYWORD ID="FILENAME"></KEYWORD>
<KEYWORD ID="MAX_WETLAND_RATIO">1.</KEYWORD>
<KEYWORD ID="MIN_WETLAND_RATIO">0.001</KEYWORD>
<KEYWORD ID="WETLAND_ID_OPTION">1</KEYWORD>
<KEYWORD ID="WI_THRESHOLD">8.</KEYWORD>
</TOPAGNPS>

```

AGWET Control File Keywords:

The following table lists the keywords that are available for the user to include in the control file. The keywords must be spelled exactly as specified below and have a comma separating the keywords. The keywords are case and order insensitive. This list is in alphabetical order, but will be described in a logical processing order later in this document.

Keyword	Description	Default
ASC_Path	This optional keyword allows the user to specify the path to where the spatial files created by DEDNM, RASPRO, RASFOR, and AgFlow are located. This path will also be used for spatial output files created by AGWET. This keyword is optional and if not included then the default location of the spatial input files and AGWET spatial output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for all of the required spatial input files and AGWET spatial output files. If the keyword is used but its value is blank, then the current path from which TopAGNPS is executed will be used for all of the spatial input files and AGWET spatial output files. NOTE: This keyword is only used for independent runs of the AGWET module. It is ignored when AGWET is run in sequence with DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from AGWET will be placed in the default folder location.	“./Output_Files/ASC”
Barrier_Height	This optional keyword allows the user to designate an amount of height in meters to be added to the elevation at the barrier-reach point and is used to define the elevation at the top of the barrier. This value is used when the “Barrier_Height_Option” is set to use a fixed barrier height value and: 1.) The barrier height values are not included in the barrier points input file or with any point that does not have an associated barrier height. 2.) The wetland identification feature (WIF) is used to produce and process barrier points. This value must be a positive real number.	1.0 [m]
Barrier_Height_Increment	This optional keyword allows the user to designate an initial and	0.1 [m]

April 24, 2023

Keyword	Description	Default
	incremental amount of height in meters to add when the barrier height is to be determined internally by virtue of the “Barrier_Height_Option” keyword defined below. This value must be a positive real number.	
Barrier_Height_Max	This optional keyword allows the user to specify a maximum barrier height limit when the barrier height is to be determined internally. Iterations determining the barrier height stop when this value is reached. This value must be a positive real number greater than the “Barrier_Height_Increment” value.	1.0 [m]
Barrier_Height_Option	This optional keyword allows the user to designate whether to use a fixed barrier height value using the “Barrier_Height” keyword or to dynamically determine the barrier height internally for all points where the barrier height value was not included in the input data file. If the “Barrier_Height” column is not included in the input data and this keyword is set to “0” then the barrier height will be determined for all points. If the “Barrier_Height” column is included in the input data then only the points without a barrier height will be determined by this process. Options: 0 = determine the barrier height internally (Blank defaults to 0) 1 = use the user-specified barrier height or global default of 1.0 meter. 2 = determine the barrier height based on the maximum elevation under the barrier line minus the barrier-reach point elevation. <i>Note: If option “0” is chosen to automatically determine the barrier height, then internally, the barrier height is set to the user-specified value by use of the “Barrier_Height_Increment” keyword and the associated wetland areas are determined. If barrier points are supplied as input then the barrier width is also determined. If the wetland ratio value, calculated as the wetland area divided by the contributing area, is less than the minimum wetland ratio then the barrier height is incrementally increased until the wetland ratio value either falls within the minimum and maximum wetland ratio limits or the wetland ratio value exceeds the upper limit. This iterative process continues until a barrier height is determined for all barrier points.</i> <i>Note: Option 2 is only used with an ASCII raster grid barrier-line input file as described in the “Wetland Barrier lines” section of this manual.</i> <i>Note: This feature may add a significant amount of execution time depending on the number of points that need a barrier height determined.</i>	0
Breach_Barrier	Allows the user to designate if the wetland extent for a given barrier based on elevation is allowed to breach another barrier and wetland extent of a lesser elevation.	0

Keyword	Description	Default
	For example, assume a reach has two barriers, one upstream of the other and breaching is allowed via this keyword. If the downstream barrier has a higher elevation than the upstream barrier and the wetland extent being determined for the downstream barrier touches the upstream barrier and wetland extent, then the downstream wetland extent will breach the upstream barrier and wetland and encompass it as one wetland. Options: 0 = allow barrier breaching. (Blank defaults to 0) 1 = no breaching allowed.	
Buffer_Extent_Option	This optional keyword allows the user to select from one of three methods used to produce the buffer extent around the wetland. Options: 0 = buffers are produced based on the flow path of flow that flows into the buffer and exits into the wetland. 1 = buffers are produced around the wetland except for downstream of the barrier. (Blank defaults to 1) 2 = buffers are produced completely encompassing the wetland.	1
Buffer_IDs_Filename	This optional keyword allows the user to designate a raster grid input file containing buffers. Each buffer must have a unique integer ID.	
Buffer_Veg_Filename	This optional keyword allows the user to designate a raster grid input file containing buffer vegetation values. Currently, AGWET only accepts the value of '1' for grass or '2' for forest.	
Buffer_Width	This optional keyword allows the user to designate if a buffer of a user-specified width in meters is to be produced around a wetland. If the value is left blank then the global default of one raster width will be used for any given barrier point that does not have a buffer width specified.	1 raster width [m]
Buffer_Zone_Filename	This optional keyword allows the user to designate a raster grid input file containing the buffer zones. Currently, AGWET only accepts a value of '2' for all rasters within the buffer zone.	
CSV_Path	This optional keyword allows the user to specify the path to where any csv-formatted input files and the csv-formatted output files created by AGWET will be written. This keyword is optional and if not included then the default location of any csv-formatted input files and AGWET csv-formatted output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for any csv-formatted input files and AGWET csv-formatted output files. If the keyword is used but its value is blank, then the current path from which TopAGNPS is executed will be used for any csv-formatted input files and AGWET csv-formatted output files. NOTE: This keyword is only used for independent runs of the AGWET module. It is ignored when AGWET is run in sequence with	“./Output_Files/ASC”

April 24, 2023

Keyword	Description	Default
	DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from AGWET will be placed in the default folder location.	
DA_Threshold	This optional keyword allows the user to designate a drainage area threshold in hectares to be used to control the number of points produced by the Wetland Identification Feature (WIF).	200 [ha]
Erosion_Index_Option	This optional keyword allows the user to designate if the LS factor data values, read from "AgFlow_LS_Factor.asc", will be used as the erosion index values or if the erosion index values will be calculated internally. This is used to control the number of points produced by the WIF. Options: 0 = use "AgFlow_LS_Factor.asc" from AgFlow (Blank defaults to 0) 1 = internally calculate the erosion index.	0
Erosion_Index_Threshold	This optional keyword allows the user to designate an erosion index threshold value to be used as a criteria to control the number of points produced by the WIF.	2.5
Filename	This optional keyword allows the user to specify the name and location of a barrier input file that is to be used for determining barriers and related wetland extents. If left blank, then the WIF is invoked. If the filename is not included then AGWET will automatically invoke the WIF to produce barrier points in two files called "AgWet_Wetland_Barrier_Generated_Points.asc" and "AgWet_Wetland_Barrier_Generated_Points.csv". "AgWet_Wetland_Barrier_Generated_Points.asc" is a raster grid file containing the barrier points and "AgWet_Wetland_Barrier_Generated_Points.csv" is a csv-formatted file that can be used as input to AGWET. Depending on the "Wetland_ID_Option" keyword, as described below, the generated points may or may not be processed to define related wetland extents. If the barrier filename is specified but not found to exist then an error message will be produced.	blank
Max_Wetland_Ratio	This optional keyword allows the user to designate the upper limit that will be used in: 1.) Selecting acceptable wetland extents for AnnAGNPS. 2.) Determining the barrier height internally.	0.02
Min_Wetland_Ratio	This optional keyword allows the user to designate the lower limit that will be used in: 1.) Selecting acceptable wetland extents for AnnAGNPS. 2.) Determining the barrier height internally.	0.005
Reach_Enhancement_Option	This optional keyword allows the user to designate if the reach ids will be segmented and reassigned according to wetlands contained within. If this option is invoked, then each original reach id will be segmented into 1-3 segments depending on the location of the wetland associated	0

Keyword	Description	Default
	within the reach.	
Wetland_ID_Option	This optional keyword allows the user to designate if the WIF is used to produce points and if those points are to be processed. This keyword, in conjunction with the "Filename" keyword, determines processing according to the matrix found in the keyword description below:	*0 or 1
WI_Threshold	This optional keyword allows the user to designate a wetness index threshold to be used to control the number of points produced by the WIF.	8.0

Required spatial layer input data files:

Seven of the required input data files are spatial layers as described below. The spatial layers must have the same dimensions with the same header information in each file. All of the filenames must be as specified. These files are automatically created as part of defining a watershed using a DEM with the TOPAGNPS modules, DEDNM, RASPRO, RASFOR, and AGFLOW. Also, if executing TopAGNPS on a UNIX/Linux system (case sensitive), the following list is in the expected case for each file.

AnnAGNPS_Cell_IDs.asc:

This required spatial input file is the cell (AGNPS cell) designations layer.

AnnAGNPS_Reach_IDs.asc:

This required spatial input file is the reach (AGNPS reach) designations layer.

BOUND.ASC:

This required spatial input file is the watershed boundary layer.

FLOVEC.ASC:

This required spatial input file is the flow vector layer.

RELIEF.ASC:

This required spatial input file is the elevation layer (m).

TSLOPE.ASC:

This required spatial input file is the slope layer (m/m).

UPAREA.ASC:

This required spatial input file is the drainage area layer.

Optional input data files:**AnnAGNPS_Cell_Data_Section.csv:**

This CSV formatted input file is produced by AgFlow. This file is only required if the “REACH_ENHANCEMENT_OPTION” keyword is used with a value of “1”. Otherwise, this file is not used by AgWet.

AgFlow_LS_Factor.asc:

This spatial input file is the RUSLE length-slope (LS) factor layer that may be used with the WIF feature. This file is optional if the “Erosion_Index_Option” keyword is specified with a value of “1” indicating that the erosion index will be calculated internally. Otherwise, this file is required.

User-specified filename (ASCII grid or comma-delimited):

This optional input file is either a csv-formatted input file (typically with a .csv extension) containing points where the wetland barrier intersects with an AnnAGNPS defined reach or a spatially-formatted wetland barrier line designation layer (with the required extension, .asc). If an input file is not specified, then the points will be automatically produced by AGWET using the criteria defined by the user or if not provided, then as defined by default parameters.

The details of this input file are described in the “Manual Wetland Identification” section of this document.

PEG_Summary.csv:

This file is produced by the PEG component of TopAGNPS.

AnnAGNPS_Ephemeral_Gully_Data_Section.csv:

This file is produced by the PEG component of TopAGNPS.

AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgBuf.csv:

This file is produced by the AgBuf component of TopAGNPS.

There are two optional csv-formatted input data files that may be used with AGWET. The two files are the output files from the Potential Ephemeral Gully (PEG) program with the following names respectively: "PEG_Summary.csv" and either "AnnAGNPS_Ephemeral_Gully_Data_Section.csv" from PEG or "AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgBuf.csv" from AgBuf. If both "AnnAGNPS_Ephemeral_Gully_Data_Section.csv" and "AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgBuf.csv" exist then "AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgBuf.csv" will be used.

If these two files exist in the folder where the required input files for AGWET reside, then they are assumed to be used as input by AGWET. If used, both must be present.

AGWET will read the list of peg points from the "PEG_Summary.csv" input file and begin processing. If a peg point is located within a wetland, then the term "WETLAND" is written to the management field ID for the gully in the "AnnAGNPS_Ephemeral_Gully_Data_Section.csv" file. AnnAGNPS will view this as indicating that the gully is ineffective and no gully sediment erosion or yield will be produced; effectively disabling the gully.

Output files:

There are several output files generated by AGWET either in the form of comma-delimited data (csv) or as spatially-formatted files (asc).

AgWet.log:

The log file, "AgWet.log", is automatically generated and is a log of the processing as the program continues its execution. Additionally, the log files for "TopAGNPS" will also contain information specific for AGWET. The log file, as it is being written, is also displayed to the screen. The log file contains processing information as well as error and warning messages.

AgWet_AnnAGNPS_Cell_Summary.csv:

This output file is a csv-formatted file that contains a summary of cell, wetland, and buffer information for any cell that contains a wetland.

AgWet_AnnAGNPS_Reach_Summary.csv:

This output file is a csv-formatted file that contains a summary of reach, wetland, and buffer information for any reach that is associated with a wetland.

AnnAGNPS_Riparian_Buffer_Data_Section_AgWet.csv:

This output file is a csv file that contains information that can be imported as the “Riparian Buffer Data” section into the AGNPS Input Editor or used directly by AnnAGNPS if used with the AnnAGNPS csv input option.

AgWet_AnnAGNPS_Wetland_Barrier_Lines.asc:

This spatially-formatted file contains the wetland barrier IDs of all barrier lines associated with each wetland included in the “AgWet_AnnAGNPS_Wetland_Data_Section.csv” output file.

AgWet_AnnAGNPS_Wetland_Barrier_Points.asc:

This output file is a spatially-formatted file that indicates the location of each wetland barrier point included in the “AgWet_AnnAGNPS_Wetland_Data_Section.csv” output file.

AgWet_AnnAGNPS_Wetland_Barrier_Points.csv:

This output file is a csv-formatted file that contains a summary of all pertinent information related to each wetland included in the “AgWet_AnnAGNPS_Wetland_Data_Section.csv” output file.

AgWet_AnnAGNPS_Wetland_Buffer_IDs.asc:

This spatially-formatted file contains the wetland buffer IDs of all buffers associated with each wetland included in the “AgWet_AnnAGNPS_Wetland_Data_Section.csv” output file.

AgWet_AnnAGNPS_Wetland_Buffer_Veg.asc:

This spatially-formatted file contains the value “1” for each raster within the buffer zone.

AgWet_AnnAGNPS_Wetland_Buffer_Zone.asc:

This spatially-formatted file contains the value “2” for each raster within the buffer zone.

AnnAGNPS_Wetland_Data_Section.csv:

This output file is formatted to be included as a “Wetland Data” section input file for AnnAGNPS.

AgWet_AnnAGNPS_Wetland_Edge.asc:

This output file is a spatially-formatted file that identifies the upstream edge rasters associated with each wetland included in the “AgWet_AnnAGNPS_Wetland_Data_Section.csv” output file.. An upstream edge raster is defined as that wetland raster where flow enters into it from a raster outside the wetland area.

AgWet_AnnAGNPS_Wetland_IDs.asc:

This output file is a spatially-formatted file that identifies all rasters for associated with each wetland included in the “AgWet_AnnAGNPS_Wetland_Data_Section.csv” output file.by their wetland ID, not including the barriers.

AgWet_Buffer_Zone_Cell_Edge.asc:

This is a computed intermediate spatial layer that assigns a value of “1”, “2”, or “3” to buffer rasters in the cells that meet certain criteria. A “1” indicates that it is an upstream-edge raster meaning that it is the first raster in the buffer along a flow-path that receives flow from outside of the buffer. A value of “2” indicates that it is a downstream-edge raster meaning that it is the last buffer raster along a flow-path before entering a reach. A value of “3” indicates that the raster is designated as both, an upstream-edge and downstream-edge raster.

AgWet_Buffer_Zone_Edge_Rasters.asc:

This is a computed intermediate spatial layer that assigns a value of “1”, “2”, or “3” to buffer rasters in the cells that meet certain criteria. A “1” indicates that it is an upstream-edge raster meaning that it is the first raster in the buffer along a flow-path that receives flow from outside of the buffer. A value of “2” indicates that it is a downstream-edge raster meaning that it is the last buffer raster along a flow-path before entering a reach that received flow from one or more upstream-edge rasters. A value of “3” indicates that the raster is a downstream-edge raster with flow that originated within the buffer; that is, there is no associated upstream-edge raster.

AgWet_Buffer_Zone_Reach_Edge.asc:

This is a computed intermediate spatial layer that assigns a value of “1”, “2”, or “3” to buffer rasters in the reaches that meet certain criteria. A “1” indicates that it is an upstream-edge raster meaning that it is the first raster in the buffer in the reach flow-path that receives flow from outside of the buffer. A value of “2” indicates that it is a downstream-edge raster meaning that it is the last buffer raster in a reach flow-path before either exiting the buffer zone or the reach. A value of “3” indicates that the raster is designated as both, an upstream-edge and downstream-edge raster.

AgWet_Calculated_Erosion_Index_Values.asc:

This is a computed spatial layer that contains internally calculated erosion index values that may be used by the WIF.

AgWet_Cell-Reach_Buffer_Flow_Path_Summary.asc:

This output file is a csv file that contains expanded information about each flow-path in the cells and reaches.

AgWet_Cell-Reach_Contributions_to_Buffers_by_Cell.asc:

Not currently produced in this release.

AgWet_Cell-Reach_Contributions_to_Wetlands_by_Cell.asc:

This output file is a csv file that contains expanded information about each cell's and reach's contribution to each wetland. This is a sorted list by cell ID order.

AgWet_Cell-Reach_Contributions_to_Wetlands_by_Wetland.asc:

This output file is a csv file that contains expanded information about each cell's and reach's contribution to each wetland. This is a sorted list by wetland ID order.

AgWet_Reach_Located_Buffers.asc:

This is a computed intermediate spatial layer that assigns a value of “1” or “2”. First, a value of “1” is assigned to all reach rasters that are located in a buffer zone. Then reach rasters that meet the criteria are assigned a value of “2”. The criteria are, 1.) the buffer must extend on both sides of the reach, and 2.) the buffer-width must meet the buffer-width threshold. This file is used to compute the “buffer_zone_reach.asc” file described above.

AgWet_Relief_Before_Adding_Wetland_Barriers.asc:

This is a computed intermediate spatial layer that contains the original raster elevations.

AgWet_Relief_Wetland_Extent.asc:

This is a computed intermediate spatial layer that contains the modified elevations showing the wetland extent including the barriers.

AgWet_Relief_With_Wetland_Barriers.asc:

This is a computed intermediate spatial layer that contains the original elevations of all but the barrier rasters. The elevation of all barriers are shown for all barrier rasters.

AgWet_Wetland_Barrier_Common_Points.csv:

This is a csv-formatted file that contains the barrier points that are common between the input file and the generated barrier points produced by the WIF. This file may be renamed and modified and then used as input back into AGWET if needed by the user. If user-defined wetland barrier points are not defined, then this file is not produced.

AgWet_Wetland_Barrier_Generated_Points.asc:

This is a computed spatial layer that contains all of the wetland generated barrier points produced by the WIF.

AgWet_Wetland_Barrier_Generated_Points.csv:

This is a computed csv-formatted file that contains all of the wetland generated barrier points produced by the WIF. This file may be renamed and modified and then used as input back into AGWET if needed by the user.

AgWet_Wetland_Barrier_Points_Area_Ratio_Criteria_Met.csv:

This output file is a csv-formatted file that contains a summary of all pertinent information related to each wetland for each barrier that met the wetland ratio criteria. In this file there may be multiple wetlands located in one reach. Currently, only one wetland is allowed per AnnAGNPS reach resulting in

these wetlands not becoming an AnnAGNPS wetland as reported in “AgWet_AnnAGNPS_Wetland_Barrier_Points.csv”.

AgWet_Wetland_Barrier_Points_Not_Used_For_AnnAGNPS.csv:

This csv-formatted output file contains information about each barrier point that was considered ineligible for AnnAGNPS. The information may be modified and used as input back into AGWET. Included in this output file is a column labeled “Reason_Code”. The column will have a value code ranging from 1 to 6 and gives an indication as to why the point was excluded as an AnnAGNPS point. The following table shows the reason code and a description:

“Reason_Code”	Description
1	An invalid barrier point was encountered in the user-specified input file. The specified “BARRIER_UTMX” or “BARRIER_UTMY” coordinate did not fall within the range of rows and columns specified in the input ASCII raster grid layers. Or An invalid barrier line was encountered in the user-specified input file. The specified line did not fall on a reach.
2*	The barrier was breached by a wetland area defined by a downstream barrier that has a higher elevation.
3	A wetland area could not be defined for the barrier. This barrier, however, was not breached by a wetland area defined by a downstream barrier with a higher elevation.
4*	A wetland area could not be defined for the barrier. This barrier was also breached by a wetland area defined by a downstream barrier with a higher elevation.
5**	The wetland area ratio did not fall within the ratio threshold criteria. The wetland area ratio is calculated as the wetland area defined for the barrier divided by the total contributing area to the wetland.
6**	The wetland area ratio fell within the ratio threshold criteria but another wetland with a larger wetland area was located on the same reach. AnnAGNPS currently only allows for one wetland per reach.
7	The wetland is ineligible because the immediately adjacent upstream reach raster has a higher elevation than the maximum barrier height and is therefore not considered to be a part of the wetland.

* Note: This can only occur if the “BREACH_BARRIER” option is set to allow breaching.

** Note: The ratio threshold criteria is not applied when a user-specified file is used for input.

AgWet_Wetness_Index_Values.asc:

This is a computed spatial layer that contains internally calculated wetness index values that will be used by the WIF to generate barrier points.

AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgWet.csv:

This output file is a csv file that potentially contains modified information for the “AnnAGNPS_Ephemeral_Gully_Data_Section.csv” file that was produced by the PEG program. This file can be imported as the “Ephemeral Gully Data” section into the AGNPS Input Editor or used directly by AnnAGNPS if used with the AnnAGNPS csv input option.

AgWet_AnnAGNPS_Reach_Summary_Revised.csv:

This output file is only produced if the “REACH_ENHANCEMENT_OPTION” is invoked. It is a csv-formatted file that contains a summary of reach, wetland, and buffer information for any reach that is associated with a wetland using the new reach ids assigned through this feature.

AnnAGNPS_Reach_Data_Section_Revised.csv:

This output file is only produced if the “REACH_ENHANCEMENT_OPTION” is invoked. It is a csv-formatted file that reflects the new reach ids assigned through this feature and is a direct input file for AnnAGNPS.

AnnAGNPS_Cell_Data_Section_Revised.csv:

This output file is only produced if the “REACH_ENHANCEMENT_OPTION” is invoked. It is a csv-formatted file that reflects the AnnAGNPS cell data section with the new reach ids assigned through this feature and is a direct input file for AnnAGNPS.

AnnAGNPS_Reach_IDs_Revised.asc:

This output file is only produced if the “REACH_ENHANCEMENT_OPTION” is invoked. It is a spatial file that reflects the new reach ids assigned through this feature.

Manual Wetland Identification:

One method of wetland identification is considered a manual process where the user supplies the filename of an input file that contains either a csv-formatted barrier points file (.csv) or a spatially-formatted wetland barrier line designations layer (.asc).

Note: When an input file is specified, all points are assumed to be considered valid and are to be included in the various output files. Therefore, no checks against any ratio threshold criteria are made even if specified in the control file.

Wetland Barrier lines:

The spatially-formatted wetland barrier line designations layer (required extension, .asc) contains rasters marked with barrier identifiers. These are unique numerical ids assigned in a straight line in the raster grid file. Multiple lines are permissible but each barrier line must have a unique numerical id > 0. Each barrier line must cross a single reach in only one place. If it does not fall on a reach, AGWET will disallow this line as a potential wetland and the line will be skipped.

The “Barrier_Height_Option” keyword is used to determine how the barrier height is determined for each barrier line. The following table shows the method of determination:

“Barrier_Height_Option”	Description
0	Barrier heights are internally determined by an iterative process as described in the “Barrier_Height_Option” keyword description.
1	Barrier heights are fixed as described in the “Barrier_Height_Option” keyword description.
2	Barrier heights are determined based on elevation as described in the “Barrier_Height_Option” keyword description.

The barrier width remains fixed as drawn in the input ASCII raster grid layer and the wetland extent is drawn based on the barrier line.

There is an optional accompanying csv-formatted file that may be used with the line-input file that gives more flexibility to the user. If used, it must be the same name as the line-input file but with a “csv” extension. For example, if the line-input file is called “Goodwin_Creek_Barrier_Lines.asc” then the optional input file must be called “Goodwin_Creek_Barrier_Lines.csv”.

The available keywords are shown in the following table:

Barrier Line File Keywords:

Keyword	Description	Default
Barrier_ID	This required keyword allows the user to designate the ID of the barrier that is to be affected by additional keywords in this table.	
Barrier_Height	This optional keyword allows the user to designate an amount of height in meters to be added to the elevation at the barrier-reach point and is used to define the elevation at the top of the barrier. This value must be a positive real number.	1.0 [m]
Barrier_Orientation	This optional keyword allows the user to designate the orientation in degrees east of north; that is North = 0 degrees. This value must be a positive real number.	Internally determined

The following example shows a sample of the contents of this optional file. The example also shows fields that are read-in and used by AGWET in processing with the exception of the “Weir_Width” and “Weir_Height”.

Required	Optional	Optional
Barrier_ID	Barrier_Height	Barrier_Orientation
9	0.9	270

Required Data:

“Barrier_ID” – The ID for the barrier line.

Optional Data:

“Barrier_Height” – This is the amount of height in meters to be added to the elevation at the barrier-reach point and is used to define the elevation at the top of the barrier. This information will be defaulted to either the user-specified default value in the control file, the global default of 1.0 meter, or the internally determined value. The default value used is based on the “Barrier_Height_Option” keyword.

“Barrier_Orientation” – This is the orientation in degrees east of north; that is North = 0 degrees. If present, this will automatically adjust to the nearest cardinal directions of 0, 45, 90, 135, 180, 225, 270, 315, or 360. If absent, the orientation will be internally determined.

Note: Multiple wetland barrier lines may fall on the same reach, but AnnAGNPS only accepts one wetland per reach. Therefore, AGWET will determine the wetland extent for all barrier lines, but where more than one barrier line falls on the same reach, only the one with the largest wetland area is considered an AnnAGNPS wetland and will be written into the “AnnAGNPS_Wetland_Data_Section.csv” output file.

Wetland Barrier points:

This input file is a csv-formatted input file (typical extension, .csv) containing wetland barrier points. The first record in this file is the header record containing keywords. The keywords must be exactly as shown below. The remaining records are related point data.

Barrier Points File Keywords:

Keyword	Description	Default
Barrier_UTMX	This required keyword allows the user to designate the UTMX coordinate for the barrier point location. This value must be a positive real number.	
Barrier_UTMY	This required keyword allows the user to designate the UTMY coordinate for the barrier point location. This value must be a positive real number.	
Barrier_Height	This optional keyword allows the user to designate an amount of height in meters to be added to the elevation at the barrier-reach point and is used to define the elevation at the top of the barrier. This value must be a positive real number.	1.0 [m]
Barrier_Orientation	This optional keyword allows the user to designate the orientation in degrees east of north; that is North = 0 degrees. This value must be a positive real number.	Internally determined
Breach_Barrier	This optional keyword allows the user to designate if the wetland extent for a given barrier based on elevation is allowed to breach another barrier and wetland extent of a lesser elevation. Options: 0 = allow barrier breaching. (Blank defaults to 0) 1 = no breaching allowed.	0
Buffer_Width	This optional keyword allows the user to designate if a buffer of a user-specified width in meters is to be produced around a wetland. If the value is left blank then the global default of one raster width will be used for any given barrier point that does not have a buffer	1 raster width [m]

Keyword	Description	Default
	width specified. If the "Buffer_Width" keyword is not found in the barrier points input file nor in the "AgWet.csv" control file then no buffer will be added around the wetlands.	
Weir_Width	This optional keyword allows the user to specify the weir width. This information is not used internally but is written to the "AnnAGNPS_Wetland_Data_Section.csv" output file	[m]
Weir_Height	This optional keyword allows the user to specify the weir height. This information is not used internally but is written to the "AnnAGNPS_Wetland_Data_Section.csv" output file. If this keyword is omitted then the barrier height will be written to the "AnnAGNPS_Wetland_Data_Section.csv" output file.	[m]

The following example shows a sample of the contents of this points file. The example also shows fields that are read-in and used by AGWET in processing with the exception of the "Weir_Width" and "Weir_Height".

Values entered for "Weir_Width" and "Weir_Height" are not used in processing but are written into the output file for AnnAGNPS called "AgWet_AnnAGNPS_Wetland_Data_Section.csv". If the "Weir_Height" is not supplied then AGWET will write the barrier height into this file.

Required	Required	Optional	Optional	Optional	Optional	Optional	Optional
Barrier_UTMX	Barrier_UTMY	Barrier_Height	Barrier_Orientation	Breach_Barrier	Buffer_Width	Weir_Width	Weir_Height
293407.83	4598312.32	0.9	270	1	15.		

Note: Several of the csv-formatted output files can be used as input back into AGWET. The information contained in these output files may vary but only the information shown in the example above will be read in and used. All other information, including the barrier and wetland id, is ignored

Required Data:

"Barrier_UTMX" & "Barrier_UTMY" – UTM coordinates for the barrier point. This point must fall on a reach as specified in the "AnnAGNPS_Reach_IDs.asc" input file. If it does not fall on a reach, AGWET will attempt to snap to the nearest reach. If successful, the X & Y coordinates of the new barrier-reach raster will be used in processing. If the snap is not successful, then the barrier point will be skipped.

Note: Multiple wetland barrier points may fall on the same reach but, AnnAGNPS only accepts one wetland per reach. Therefore, AGWET will determine the wetland extent for all wetland barrier points but where more than one fall on the same reach, only the point with the largest wetland area is considered an AnnAGNPS wetland and will be written into the "AnnAGNPS_Wetland_Data_Section.csv" output file.

Optional Data:

"Barrier_Height" – This is the amount of height in meters to be added to the elevation at the barrier-reach point and is used to define the elevation at the top of the barrier. This information will be defaulted to either the user-specified default value in the control file, the global default of 1.0 meter, or the internally determined value. The default value used is based on the "Barrier_Height_Option" keyword.

“Barrier_Orientation” – This is the orientation in degrees east of north; that is North = 0 degrees. If present, this will automatically adjust to the nearest cardinal directions of 0, 45, 90, 135, 180, 225, 270, 315, or 360. If absent, the orientation will be determined as the cardinal perpendicular to the angle calculated by using two points; 1. The barrier-reach point and 2. Upstream 5th reach raster or downstream 5th reach raster if the barrier point is at the upstream end of the reach.

“Breach_Barrier” – This allows the user to designate if the individual wetland extent being determined based on elevation is allowed to breach another barrier and wetland extent of a lesser elevation.

For example, assume a reach has two barriers, one upstream of the other and breaching is allowed. If the downstream barrier has a higher elevation than the upstream barrier and the wetland extent being determined for the downstream barrier touches the upstream barrier and wetland extent, then the downstream wetland extent will breach the upstream barrier and wetland and encompass it as one wetland with the downstream barrier identification.

“Buffer_Width” – This is the buffer width in meters to be added around each wetland. This information will be defaulted to either the user-specified default value in the control file or the global default of 1 raster width in meters. The default value of 1 raster width is equal to the cellsize specified in the “cellsize” record found in the ASCII raster grid input files. If the “Buffer_Width” keyword is not found in the barrier points input file nor in the “AgWet.csv” control file then no buffer will be added around the wetlands.

“Weir_Width” – This information is not used internally but is passed through into the “AnnAGNPS_Wetland_Data_Section.csv” output file.

“Weir_Height” – This information is not used internally but is passed through into the “AnnAGNPS_Wetland_Data_Section.csv” output file.

Automatic Wetland Identification:

The other method of wetland identification is considered an automated process where the user does not supply the input filename for wetland barrier points or lines. AGWET will determine a list of barrier points based on the keywords in the control file using the “Wetland Identification Feature” (WIF).

AGWET Control File Keywords:

The following keywords may be used to affect the number of points that AGWET produces. The keywords must be spelled exactly as specified below and have a comma separating the keywords. The keywords are case and order insensitive. The list below is ordered in a more logical sequence of expected use rather than alphabetical.

“Filename”

Leave blank to invoke the automated WIF.

“Breach_Barrier”

Allows the user to designate if the wetland extent for a given barrier based on elevation is allowed to breach another barrier and wetland extent of a lesser elevation.

For example, assume a reach has two barriers, one upstream of the other and breaching is allowed via this keyword. If the downstream barrier has a higher elevation than the upstream barrier and the wetland extent being determined for the downstream barrier touches the upstream barrier and wetland extent, then the downstream wetland extent will breach the upstream barrier and wetland and encompass it as one wetland.

Options: 0 = allow barrier breaching. (Blank defaults to 0)
 1 = no breaching allowed.

“Barrier_Height_Option”

This optional keyword allows the user to designate whether to use a fixed barrier height value using the “Barrier_Height” keyword or to dynamically determine the barrier height internally for all points where the barrier height value was not included in the input data file.

If the option to automatically determine the barrier height is chosen, then internally, the barrier height is set to the user-specified value by use of the “Barrier_Height_Increment” keyword and the barrier width and the associated wetland area are determined. If the wetland ratio value, calculated as the wetland area divided by the contributing area, is less than the minimum wetland ratio then the barrier height is incrementally increased until the wetland ratio value either falls within the minimum and maximum wetland ratio limits or the wetland ratio value exceeds the upper limit. This iterative process continues until a barrier height is determined for all barrier points.

If the “Barrier_Height” column is not included in the input data and this keyword is set to “0” then the barrier height will be determined for all points. If the “Barrier_Height” column is included in the input data then only the points without a barrier height will be determined by this process.

Options: 0 = determine the barrier height internally (Blank defaults to 0)
 1 = use the user-specified barrier height or global default of 1.0 meter.

Note: This feature may add a significant amount of execution time depending on the number of points that need a barrier height determined.

“Barrier_Height”

This optional keyword allows the user to designate an amount of height in meters to be added to the elevation at the barrier-reach point and is used to define the elevation at the top of the barrier.

This value is used when the “Barrier_Height_Option” is set to use a fixed barrier height value and the wetland identification feature (WIF) is used to produce and process barrier points. This value must be a positive real number.

Default = 1.0 [m]

“Barrier_Height_Increment”

This optional keyword allows the user to designate an initial and incremental amount of height in meters to add when the barrier height is to be determined internally by virtue of the “Barrier_Height_Option” keyword defined below. This value must be a positive real number.

Default = 0.1 [m]

“Barrier_Height_Max”

This optional keyword allows the user to specify a maximum barrier height limit when the barrier height is to be determined internally. Iterations determining the barrier height stop when this value is reached. This value must be a positive real number greater than the “Barrier_Height_Increment” value.

Default = 1.0 [m]

“Wetland_ID_Option”

This optional keyword allows the user to designate if the WIF is used to produce points and if those points are to be processed. This keyword, in conjunction with the “Filename” keyword, determines processing according to the following matrix:

Filename	Wetland_ID_Option*	Results
Blank	0	WIF points produced but not processed
Blank	1	WIF points produced and processed
Present	0	Input file read and processed; WIF points not produced
Present	1	Input file read and processed; WIF points produced but not processed

**Note: The default value for this keyword is set according to the following:*

Default = 1, if the "Wetland_ID_Option" keyword is included in the control file but its value is left blank.

Default = 1, if the "Wetland_ID_Option" keyword is not included in the control file but the "DA_Threshold", "Erosion_Index_Threshold", and "WI_Threshold" keywords are included.

Default = 0, if the "Wetland_ID_Option" keyword is not included in the control file nor are the "DA_Threshold", "Erosion_Index_Threshold", and "WI_Threshold" keywords.

"Reach_Enhancement_Option"

This optional keyword allows the user to designate if the reach network should be further segmented based on the location of associated wetlands. If invoked, the original reach IDs will be segmented and new IDs assigned based on the following:

First, all original reach IDs are multiplied * 10 forming the basis of new reach ids. Then each reach that touches a wetland is divided into up to 3 segments. A value of 1, 2, or 3 is then added to the new reach id where, "3" = upstream segment above the wetland; "2" = reach segment inside the wetland; "1" = reach segment downstream of the wetland.

For example:

Assume the original reach id "3" has no associated wetland, then its new reach id = "30".

Assume the original reach id "12" has a wetland in the middle of the reach, then its "up" segment has an id of $12 * 10 + 3$ or "123"; its "wet" segment has an id of $12 * 10 + 2$ or "122"; its "down" segment has an id of $12 * 10 + 1$ or "121".

Therefore, the last digit of the new reach id allows for quick identification. If it ends in "0", then it has no associated wetland. If it ends in "1" then it is the "down" segment after leaving a wetland. If it ends in "2" then it is the "wet" segment. If it ends in "3", then it is the "up" segment prior to entering a wetland.

Some common variations of reach ID reassignment are (last digit of new reach id):

Some reaches do not have a wetland associated with it. (0)

Some reaches are entirely encompassed within a wetland. (2)

Some reaches have upstream end in a wetland but not the downstream end. (2,1)

Some reaches have downstream end in a wetland but not the upstream end. (3,2)

Some reaches have a wetland in the middle but does not extend to the upstream end nor to the downstream end. (3,2,1)

Options: 0 = Do not segment reaches based on wetland placement. (Blank defaults to 0)
 1 = segment reaches based on wetland placement.

"Erosion_Index_Option"

April 24, 2023

This optional keyword allows the user to designate if the LS factor data values, read from “AgFlow_LS_Factor.asc”, will be used as the erosion index values or if the erosion index values will be calculated internally. This is used to control the number of points produced by the WIF.

Options: 0 = use “AgFlow_LS_Factor.asc” from AgFlow (Blank defaults to 0)
 1 = internally calculate the erosion index.

“Erosion_Index_Threshold”

This optional keyword allows the user to designate an erosion index threshold value to be used as a criteria to control the number of points produced by the WIF.

Default = 2.5

“DA_Threshold”

This optional keyword allows the user to designate a drainage area threshold in hectares to be used to control the number of points produced by the Wetland Identification Feature (WIF).

Default = 200.0 [ha]

“WI_Threshold”

This optional keyword allows the user to designate a wetness index threshold to be used to control the number of points produced by the WIF.

Default = 8.0

“Min_Wetland_Ratio”

This optional keyword allows the user to designate the lower limit that will be used in:

- 1.) Selecting acceptable wetland extents for AnnAGNPS.
- 2.) Determining the barrier height internally.

Default = 0.005 (0.5%)

“Max_Wetland_Ratio”

This optional keyword allows the user to designate the upper limit that will be used in:

- 1.) Selecting acceptable wetland extents for AnnAGNPS.

April 24, 2023

2.) Determining the barrier height internally.

Default = 0.02 (2%)